

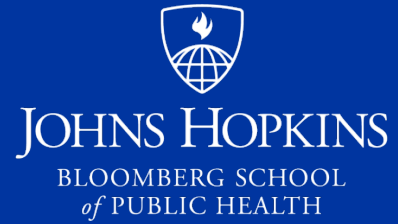
# Problem Solving Methodology

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Keri Althoff, PhD  
Johns Hopkins University



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## Introduction to the Methodology

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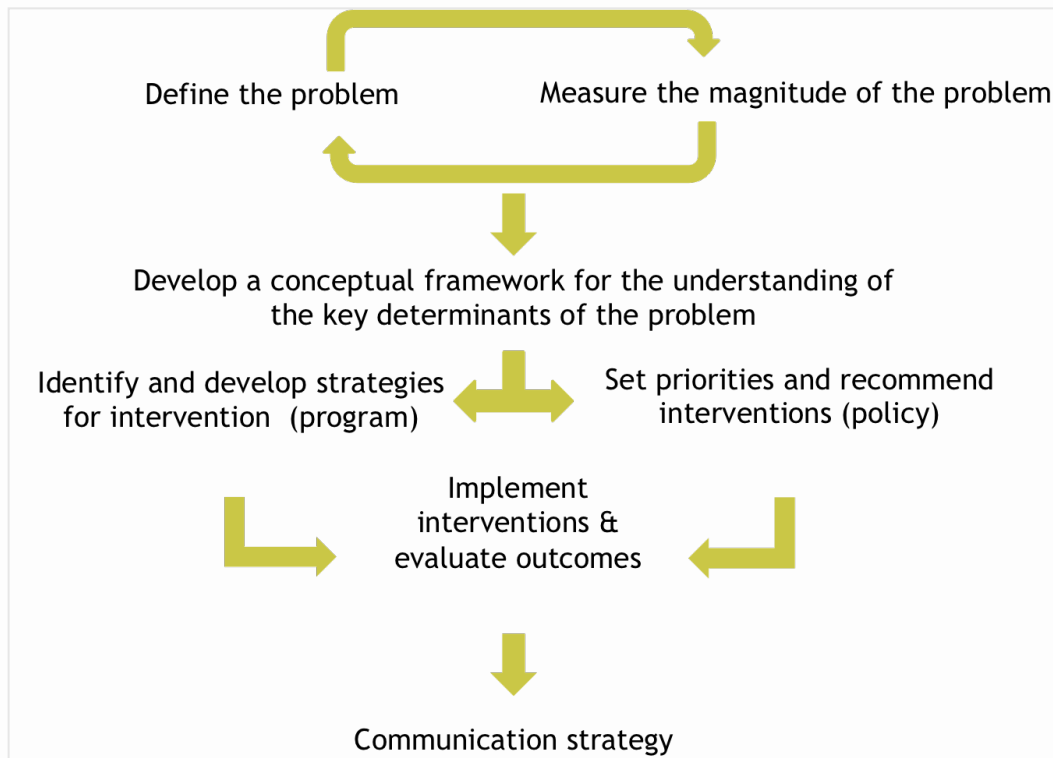
## Supporting Readings

- Guyer B (1998). Problem-solving in public health. In H. Armenian and S. Shapiro, (Eds) *Epidemiology and Health Services Research*. New York: Oxford University Press.

## Objectives

- To describe the steps in the problem solving methodology (PSM) with application to a population-based health problem
- **After this lecture, you should be able to:**
  - ▶ Apply the problem solving methodology to a public health problem of interest to you

# Problem Solving Methodology



- Professional Epidemiologist plays a role in each step of this process, but it takes a team to effectively solve public health problems!

## The Problem

- *The Baltimore Sun* newspaper reports the infant mortality rate (IMR) in Baltimore City is much higher than the state of Maryland, higher than the US, and comparable to the IMR in low- and middle-income countries
- The Mayor calls the Health Commissioner and says “do something”
- The Health Commissioner tells the Chief Epidemiologist to start investigating the problem and what can be done
- You are in the Epi Bureau at the Baltimore City Health Department (BCHD) and the Chief Epidemiologist tasks you with the Health Commissioner’s request

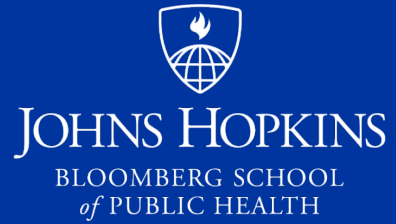
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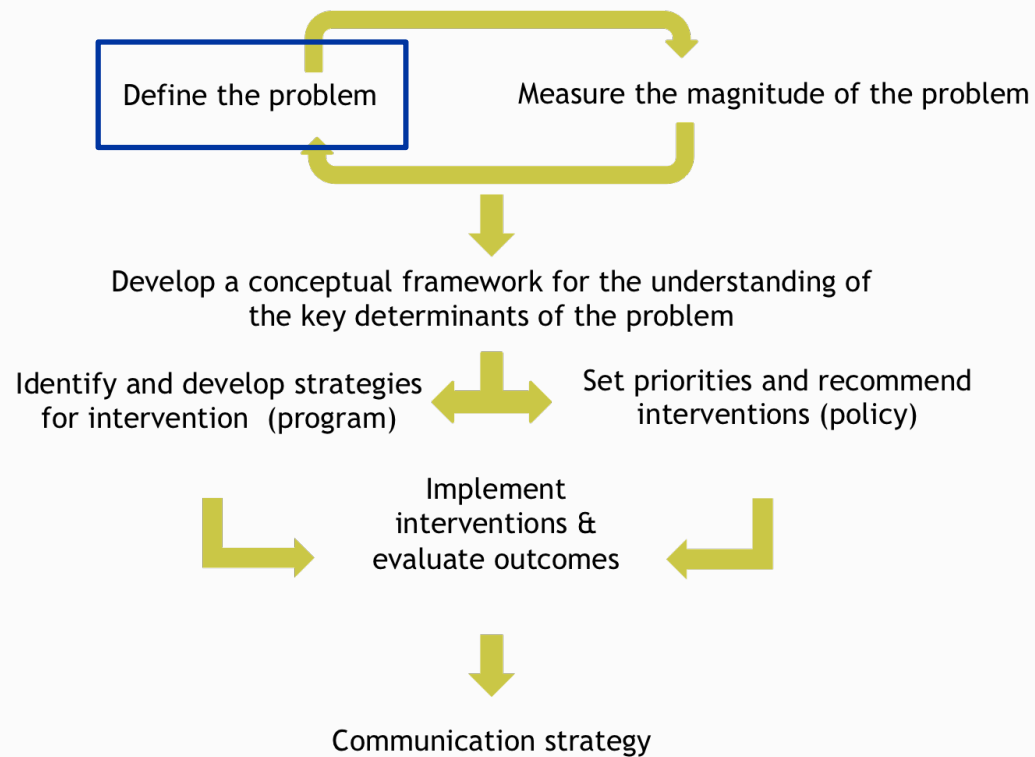
## Problem Definition Part 1

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*“A problem well stated is a problem half solved”  
~John Dewey (1859-1952), philosopher*

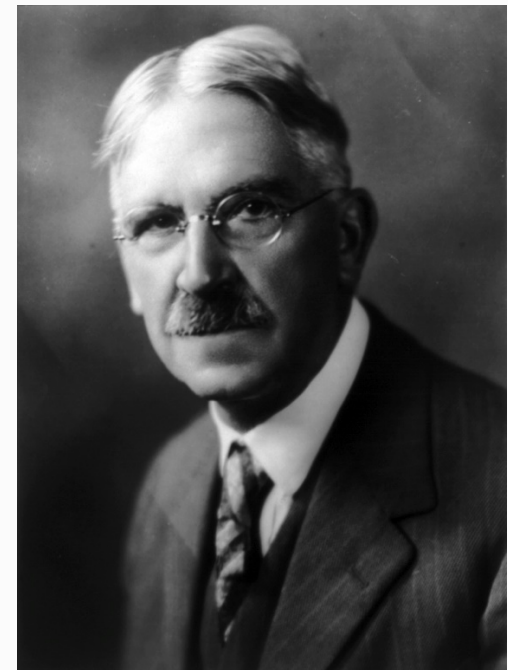


# Problem Solving Methodology



# Problem Definition

- Creating a problem is an iterative process
  - ▶ Non-linear process
  - ▶ Changes in the definition are expected
  - ▶ Requires informed discussion
  - ▶ Tests team skills
  
- Defining a problem is an iterative process and is often the most difficult step in the problem solving methodology, but a good problem definition goes a long way towards successfully solving the problem



## Key Aspects of a Problem Statement

- Nature of the problem/focus
- Framing of the problem
  - ▶ Person
  - ▶ Place
  - ▶ Time
- Specificity of the problem
- Causality
- Language considerations
- Do's and Don'ts

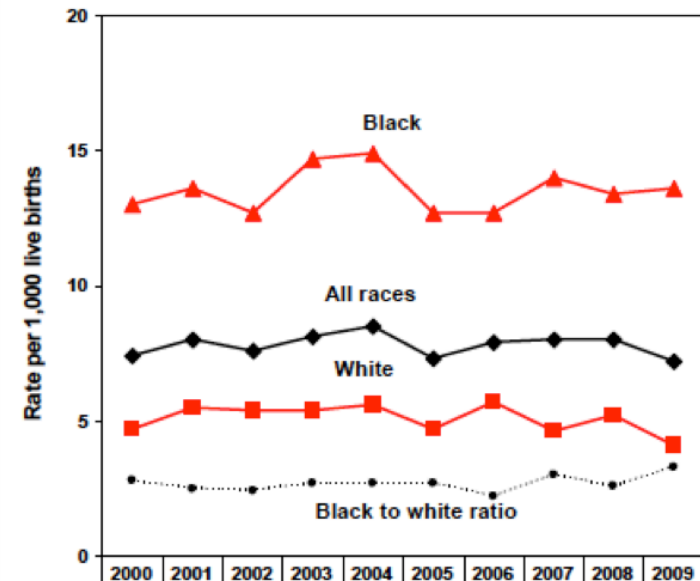
## Nature of the Problem / Focus

- Is this a focused public health problem?
- Not all health problems are public health problems
  - ▶ Clinical patient management vs. public health
- If you are too broad
  - ▶ Problem might begin to cross the line into other fields such as economic or political problems

## Framing of the Problem - Person

- Who is affected by this problem?
- Be as specific as possible...
  - ▶ ... using basic demographics
    - Gender
    - Age
    - Race/ethnicity
    - Socioeconomic status (education, income, etc)
  - ▶ ... and using problem-specific demographics
    - Sexual orientation
    - IDU
    - Housing situation
    - Occupation

Figure A. Infant Mortality Rates by Race and Black to White Ratio, Maryland, 2000-2009.



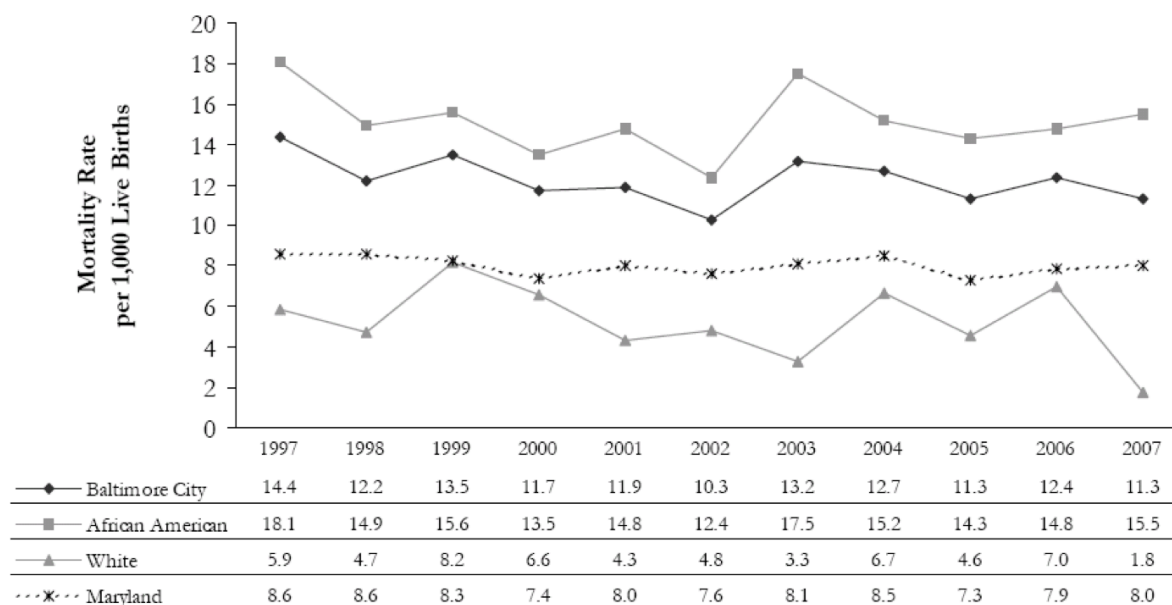
Maryland Vital Statistics, Maryland State Infant Mortality Report, 2009. July 2010. Available at <http://www.healthybabiesbaltimore.com/uploads/file/Maryland%20State%20IMR%20report%202009.pdf>

## Framing of the Problem - Place 1

- Where is the problem located?
- Be as specific as possible
  - ▶ Geographical
  - ▶ Cultural
  - ▶ Social
  - ▶ Political
  - ▶ Economic
  - ▶ Historical
  - ▶ Ethnic

## Framing of the Problem - Place 2

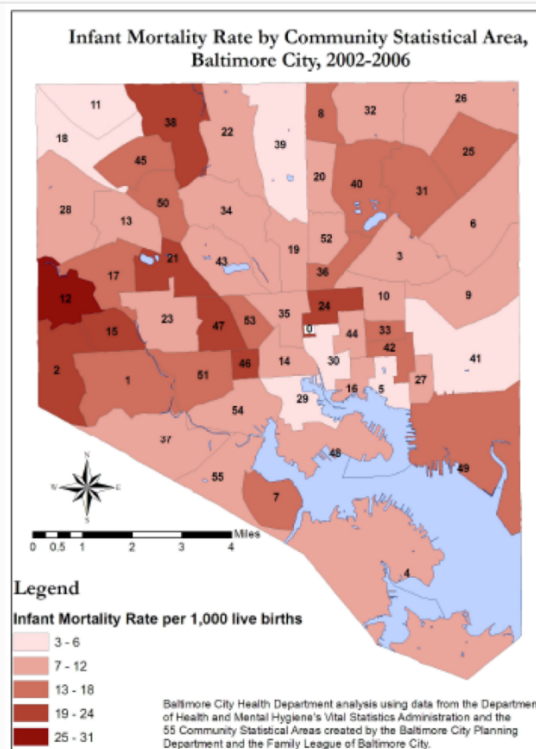
**Infant Mortality Rate, Baltimore City by Race and Maryland, 1997-2007**



Source: Maryland Department of Health and Mental Hygiene, Maryland Annual Vital Statistics Report (2007 data are preliminary).  
Rates for other races and ethnicities are not available due to the small number of events in each subgroup.

Maternal and Infant Health: Infant Mortality. Baltimore City Health Status Report, 2008. Baltimore City Department of Health.  
<http://www.baltimorehealth.org/info/HSR/InfantMortality.pdf>

# Infant Mortality Rate in Baltimore City



Source: Baltimore City Health Department

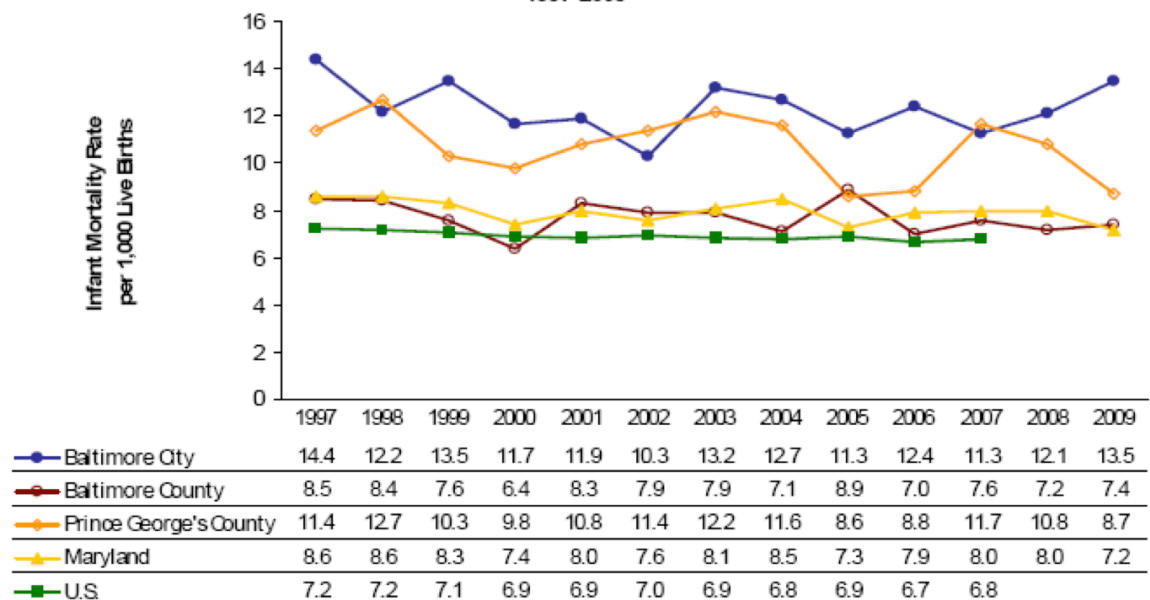


## Framing of the Problem - Time

- Where is the problem located?
- Be as specific as possible:
  - ▶ Geographical
  - ▶ Cultural
  - ▶ Social
  - ▶ Political
  - ▶ Economic
  - ▶ Historical
  - ▶ Ethnic

# Infant Mortality Rate in Baltimore City

Infant Mortality Rate, Baltimore City, Maryland State and Select Counties and U.S., 1997-2009



Source - Maryland State and County Data: Maryland Department of Health and Mental Hygiene, Maryland Annual Vital Statistics Reports 1997-2009. U.S. Data: Xu J. et al. Deaths: Final data for 2007. National vital statistics reports; vol 58 no 19. Hyattsville, Maryland: National Center for Health Statistics. 2010. U.S. data for 2008 and 2009 are not yet available.

## Specificity of the Problem

- What is the *specific* dimension or aspect of interest?
- If you think you are being too specific, you probably are **not**
- Avoid broad, general topics as they run the risk of containing competing concepts

## Causal Claims

- Are etiologic factors or causes part of your problem definition?
- **AVOID** causal claims
- Be short and clear - this is the problem definition, not the answer to the problem

## Language Considerations

- Be clear and precise
- Define and reference technical terms
- Define local terms
- Make sure it is a statement
  - ▶ Grammar, sentence structure, etc.

## Do's and Don'ts

- **Don't** state the definition of a topic (as opposed to a problem)
- **Do** be clear in defining the population or sub-population
- **Don't** mention determinants (that are not associated with person, place, or time) or make causal claims in the problem definition
- **Don't** suggest a solution in the problem statement

## Draft the problem definition

Work it out >>

Write a problem definition for infant mortality in Baltimore City.

Writing problem statements takes practice, practice, and more practice!



# Problem Solving Methodology

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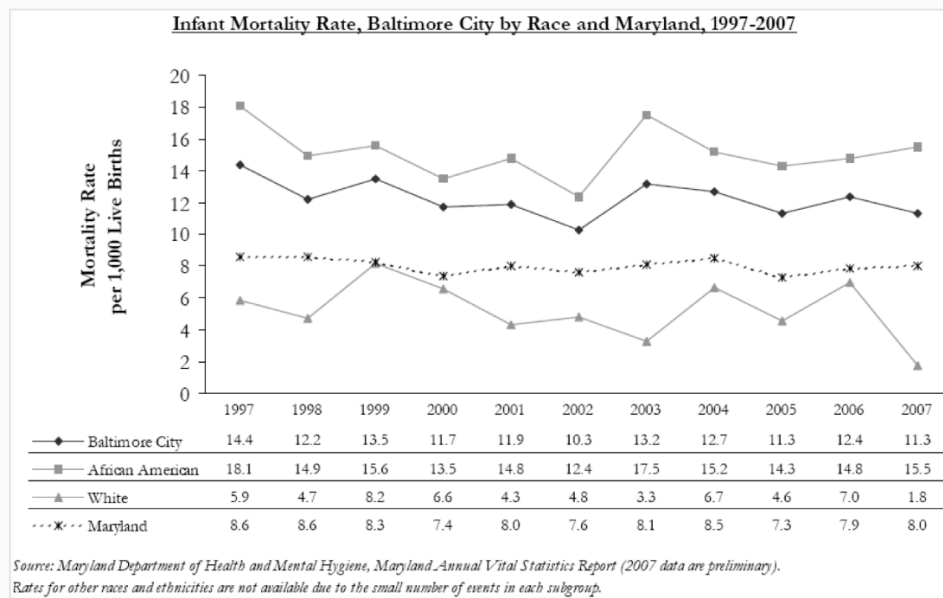
## Problem Definition Part 2

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*“A problem well stated is a problem half solved”  
~John Dewey (1859-1952), philosopher*

## What is our initial problem statement?

Infant mortality rate among African American infants was consistently higher than among White infants in Baltimore City from 1997 to 2007.



## Revise your problem definition

**WORK IT OUT >>**

**Critically review your problem definition for  
infant mortality in Baltimore City  
and revise.**

Writing problem statements takes practice, practice, and more practice!



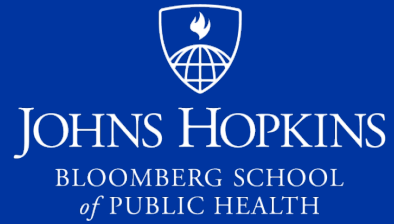
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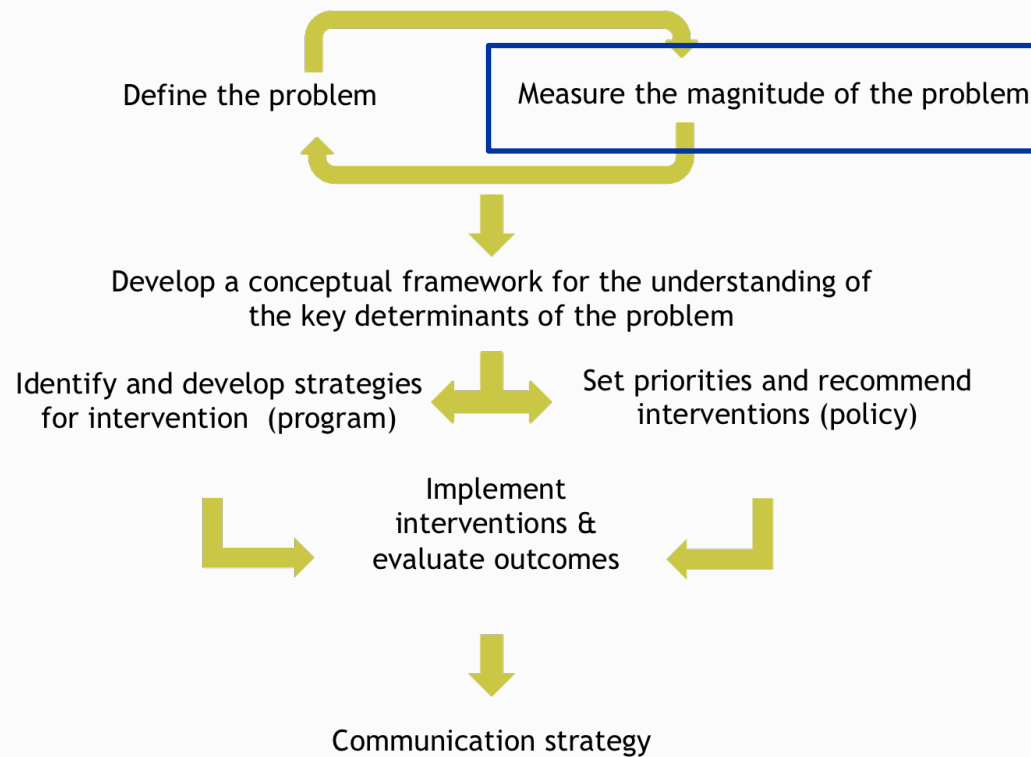


## Magnitude of the Problem

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*“A problem well stated is a problem half solved”  
~John Dewey (1859-1952), philosopher*

# Problem Solving Methodology



## Measuring the Magnitude

- IMR is regularly measured data from what vital statistics data
- Are we done with this step? Isn't the problem already measured?
  - ▶ There are always more ways to more specifically measure the magnitude!



## Other ways to Describe the Magnitude

- In 2009
  - ▶ 128 (**number**) infants died in Baltimore City (13.5/1,000 live births)
  - ▶ >10 babies/month on **average**
  - ▶ Among major US cities, Baltimore has the fourth highest infant mortality rate (**ranking**)
  - ▶ **Racial disparities**
    - African American infants: IMR = 18.5/1,000 live births (66% of births in Baltimore are to African American mothers)
    - White infants: IMR = 3.5/1,000 live births
- IMR has not improved over the past half decade (**time**)





# Problem Solving Methodology

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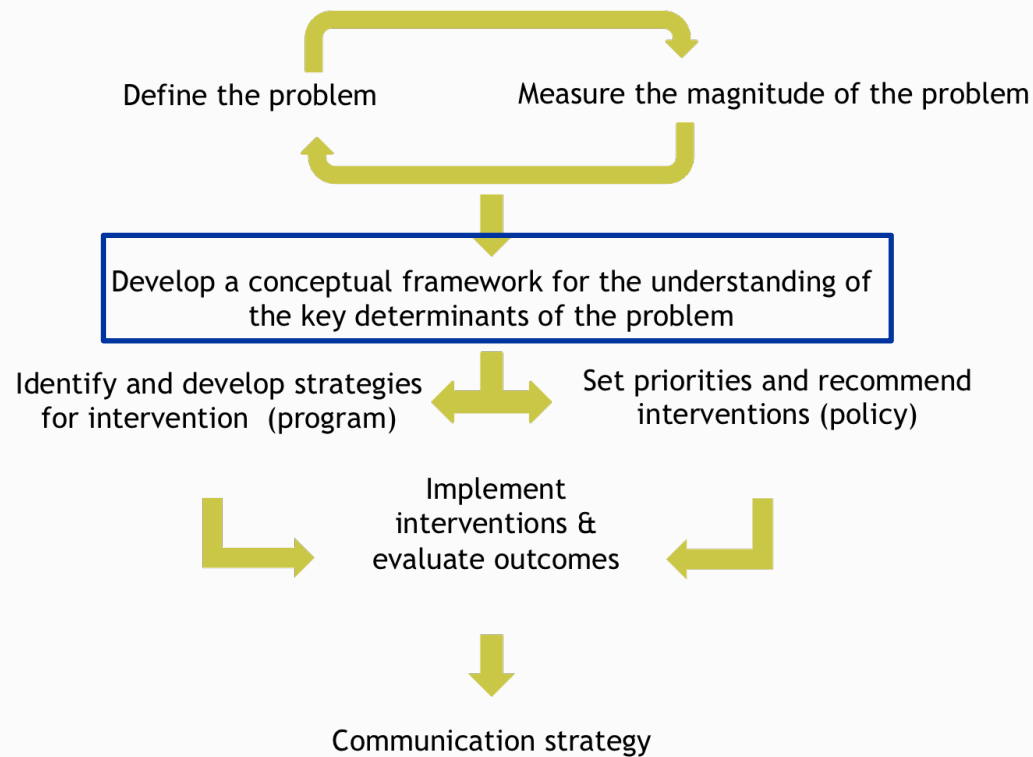
## Conceptual Framework

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“A conceptual framework is a ‘frame that works’  
to put those concepts into practice.”

*~Paul Hughes, Ten Meters of Thinking*

# Problem Solving Methodology



## Building a Conceptual Framework

- Sources of information to build a conceptual framework
  - ▶ Peer-reviewed literature
  - ▶ Grey Literature
  - ▶ Consensus Statements
- Indicators
  - ▶ Measures of key determinants
  - ▶ Measure of outcome

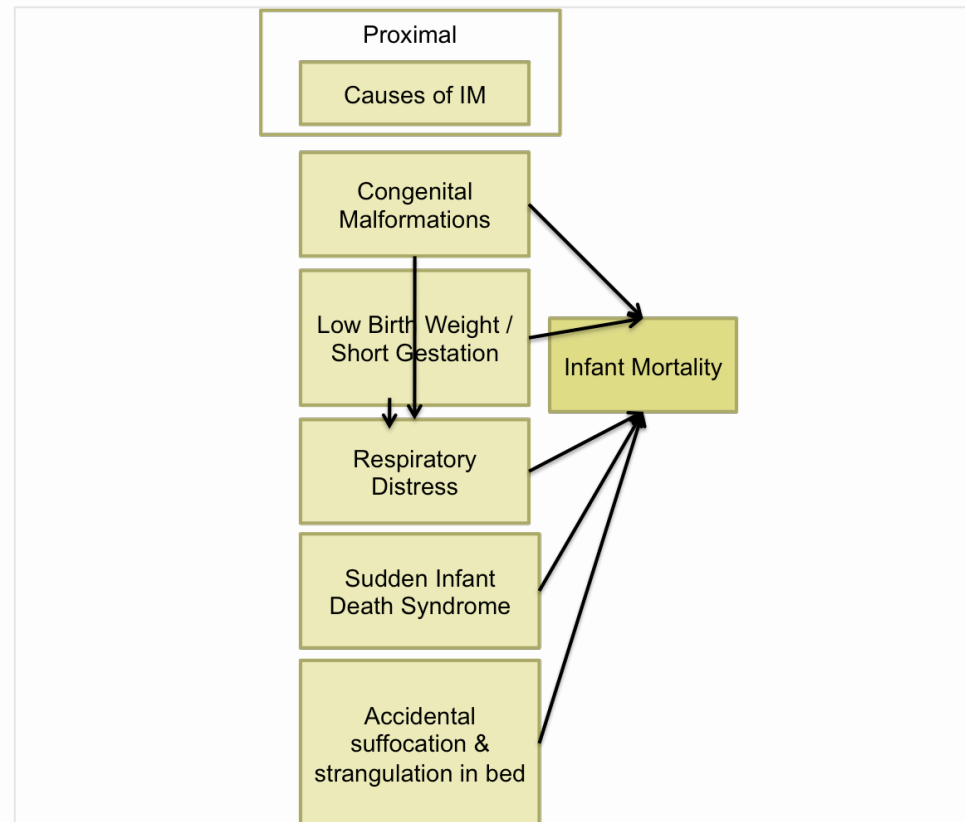
## Tools to build a conceptual framework: Proximal Determinants

- Closest to the outcome
- 2 categories for the framework we will be using
  - ▶ Direct causes of the outcome
  - ▶ Individual characteristics that may or may not be causal
- Often the determinant in the framework on which one will intervene is located under proximal determinants
  - ▶ Informs “HOW we intervene”
  - ▶ Intervening on proximal determinants typically results in a greater decrease of the problem

## Tools to build a conceptual framework: Distal Determinants

- Further removed for the outcome
- Often these determinants are ASSOCIATED with the outcome, not causally related
- 2 categories for the framework we will be using
  - ▶ Environmental
  - ▶ Non-modifiable
    - Individual characteristics that will most likely not be modifiable with an intervention that is specific to the public health problem (i.e., low educational status of the mother in the problem of infant mortality)
    - Non-modifiable characteristics often inform “to WHOM should we target the intervention?”

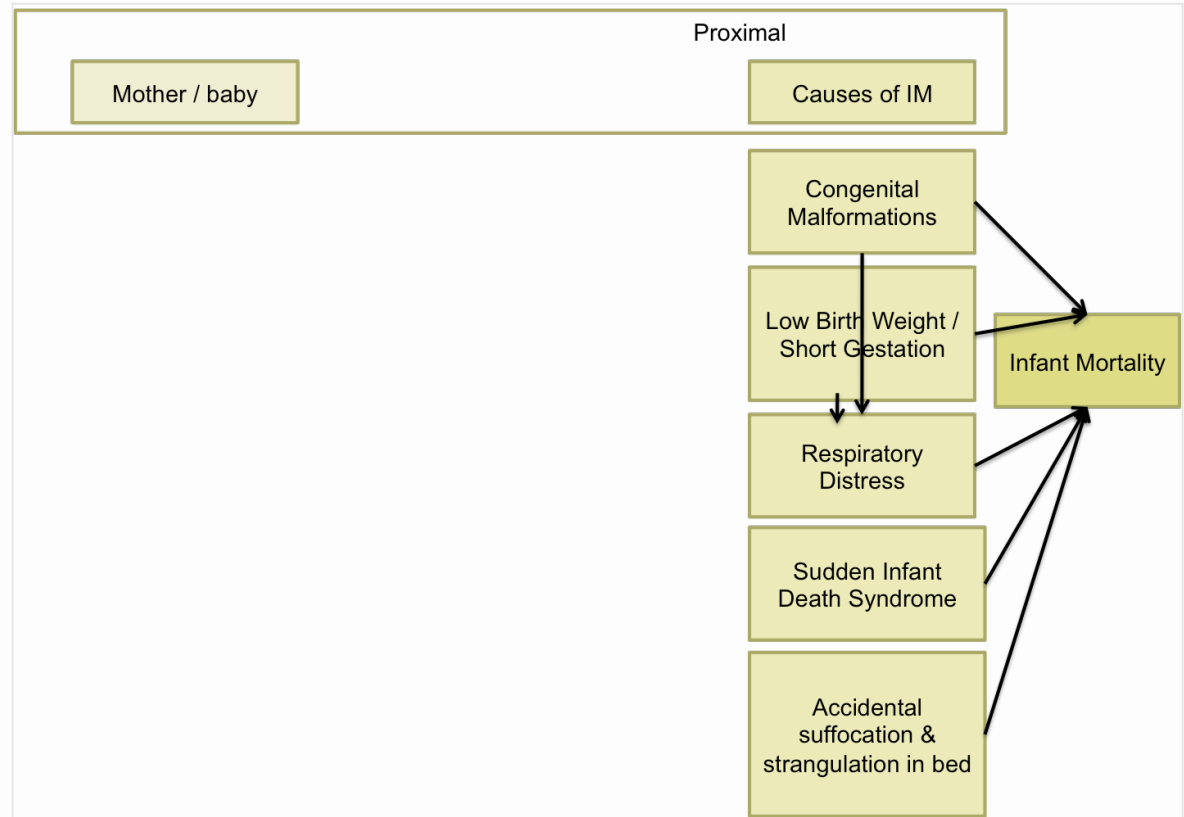
## Building a Conceptual Framework 1



## Building a Conceptual Framework 2

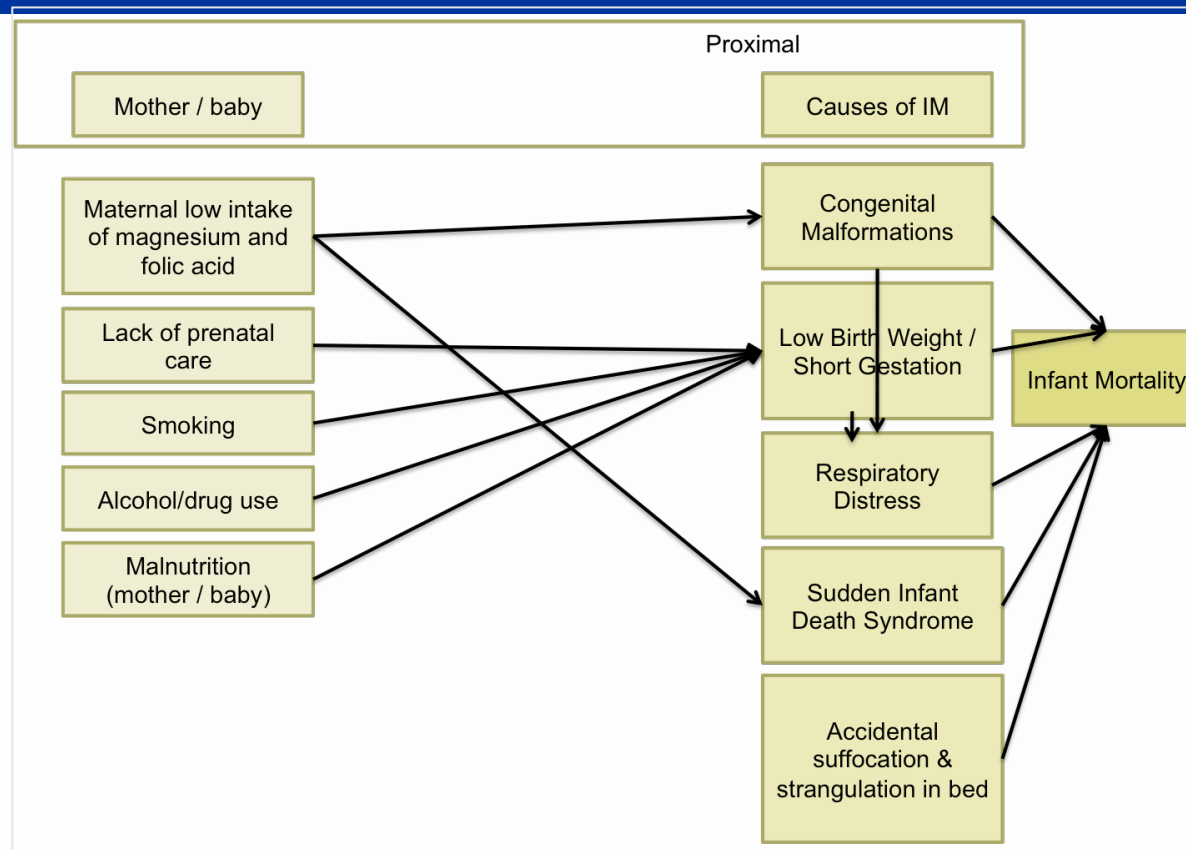
### WORK IT OUT >>

- Take a moment to add proximal determinants of the mother and baby to the conceptual framework





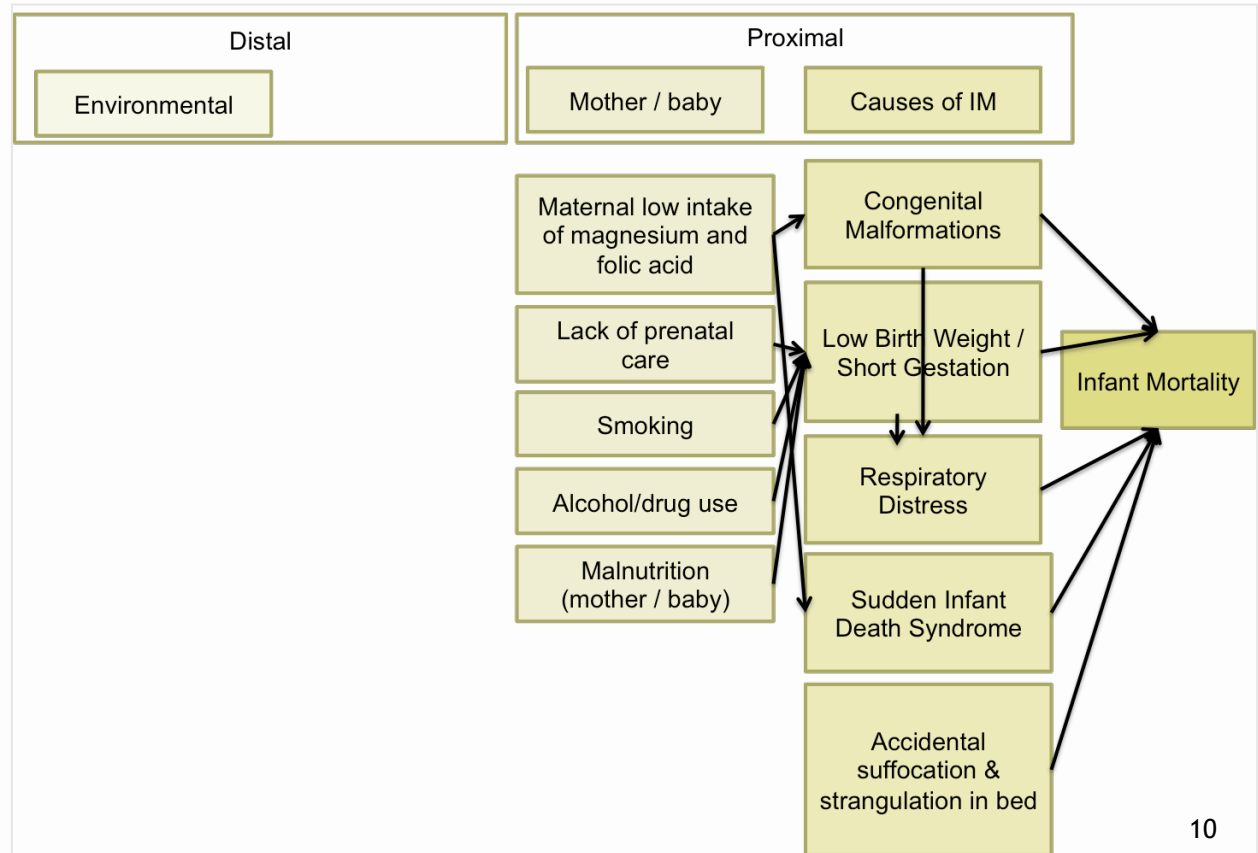
## Building a Conceptual Framework 3



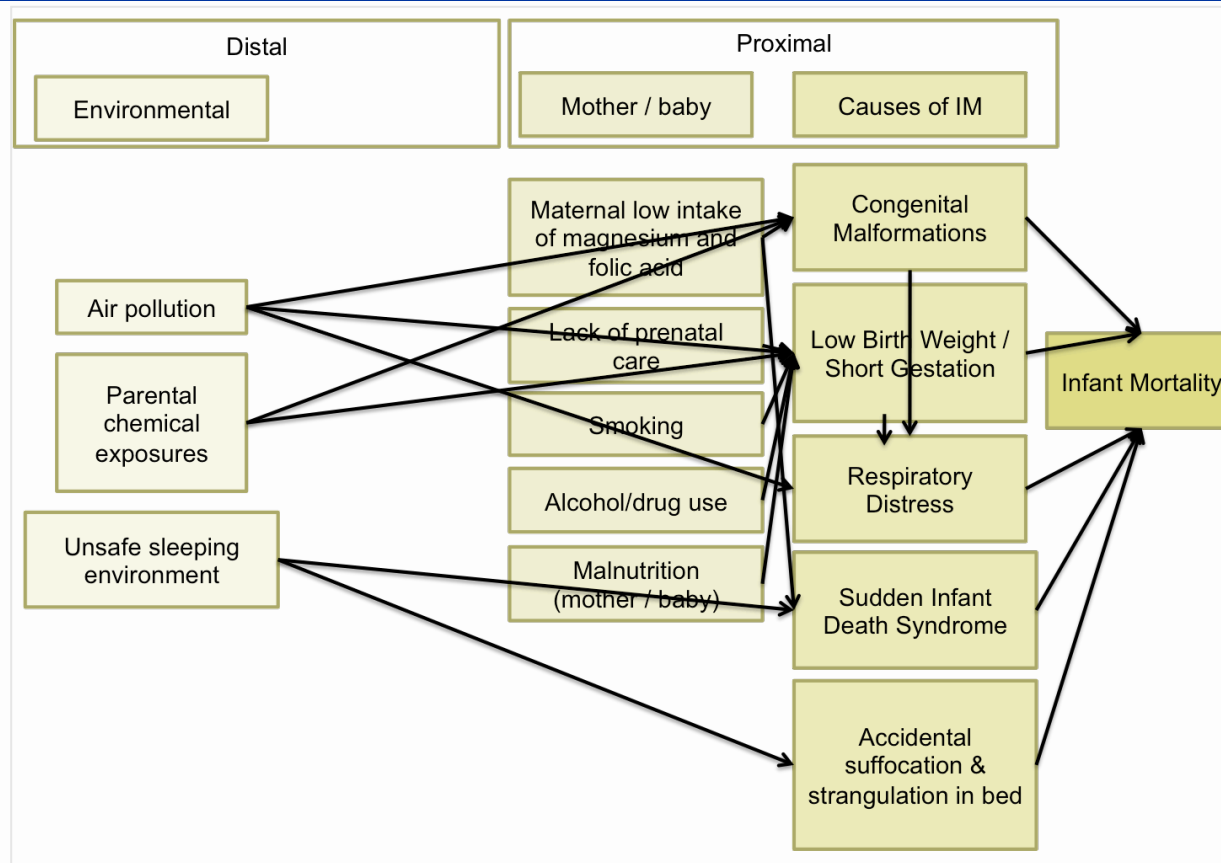
## Building a Conceptual Framework 4

### WORK IT OUT >>

- Take a moment to add distal environmental determinants to the conceptual framework



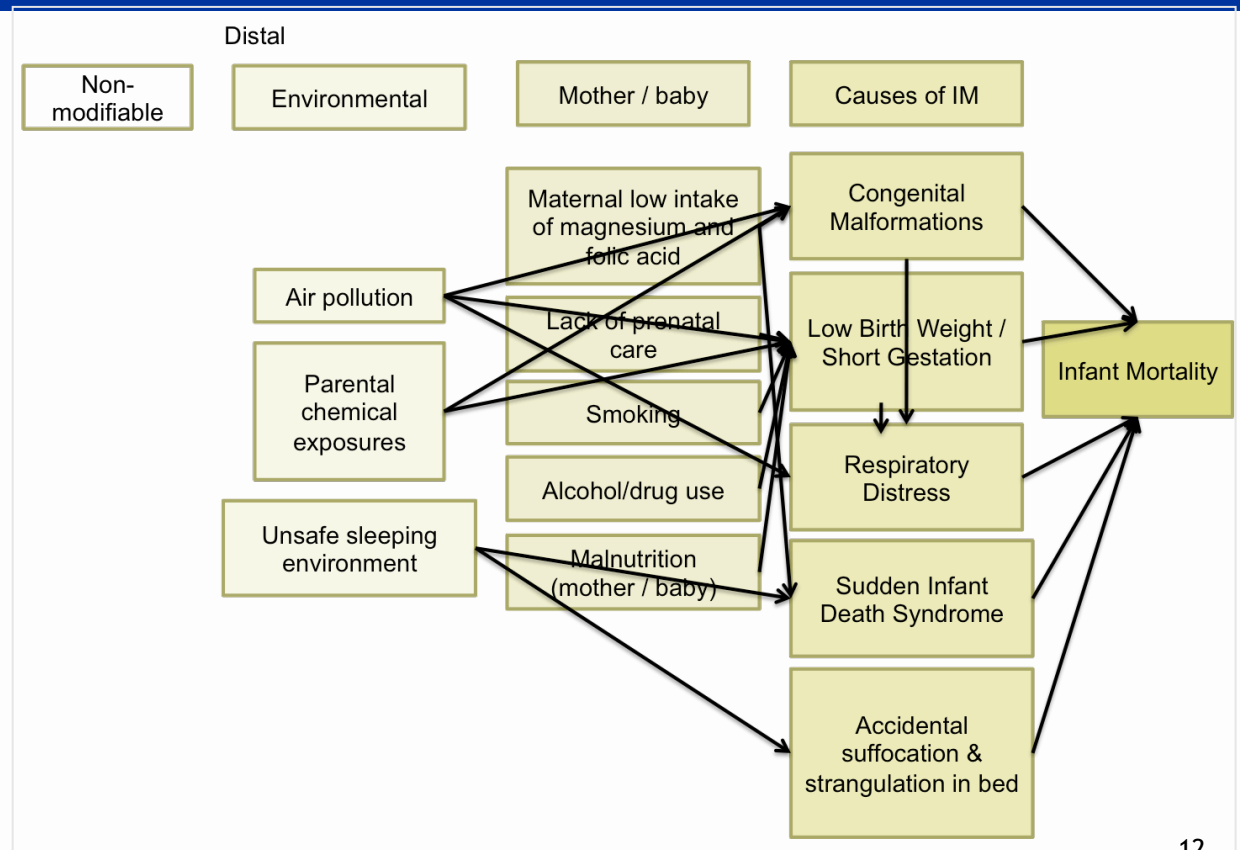
## Building a Conceptual Framework 5



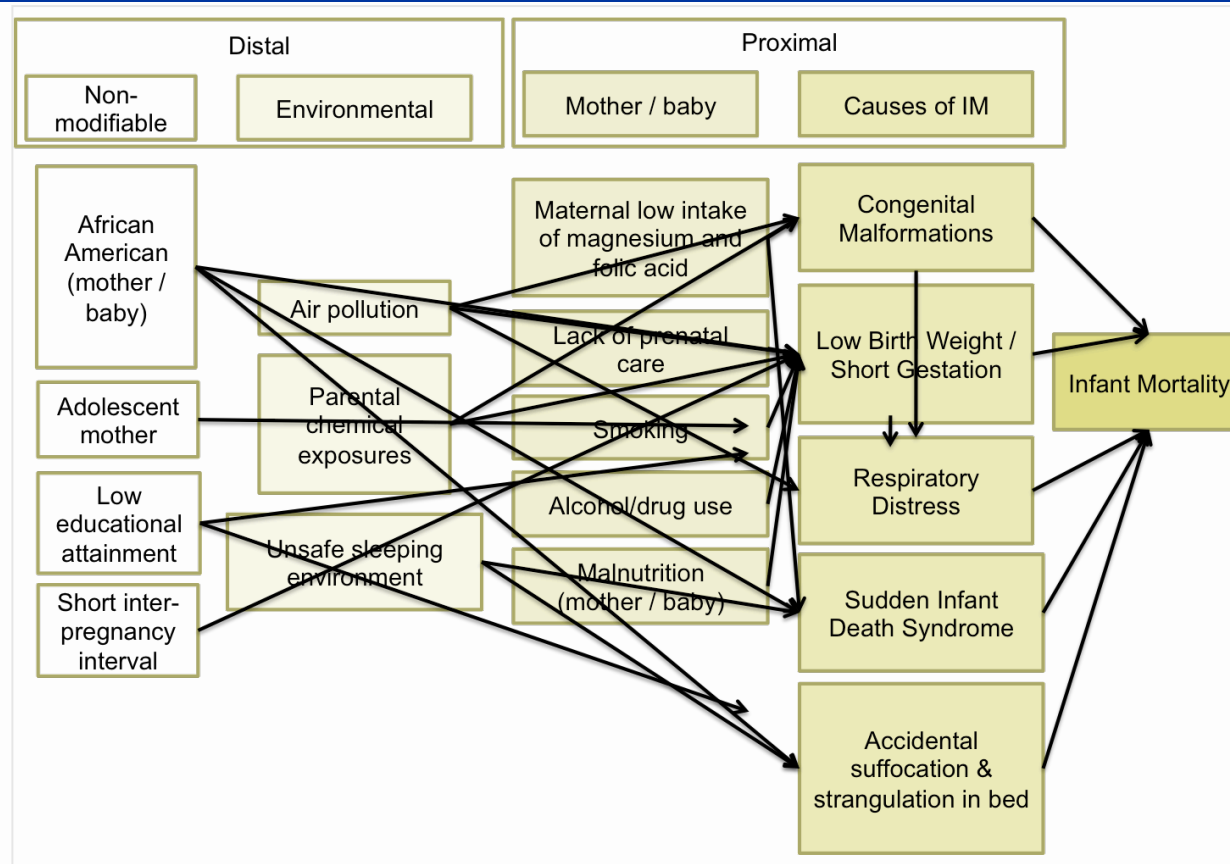
## Building a Conceptual Framework 6

### WORK IT OUT >>

- Take a moment to add distal non-modifiable factors to the conceptual framework



## Building a Conceptual Framework 7



## Two main uses of the conceptual framework

- To identify the target group for the intervention - the WHO
  - ▶ Individual characteristics
  - ▶ Non-modifiable characteristics
  
- To identify how to intervene - the HOW
  - ▶ Usually intervening on proximal causes creates a greater chance at seeing the intended impact of the intervention



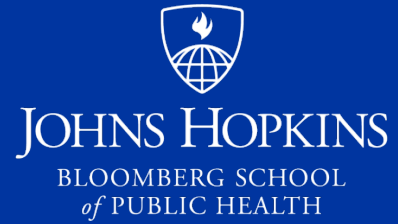
# Problem Solving Methodology

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## Action Plan

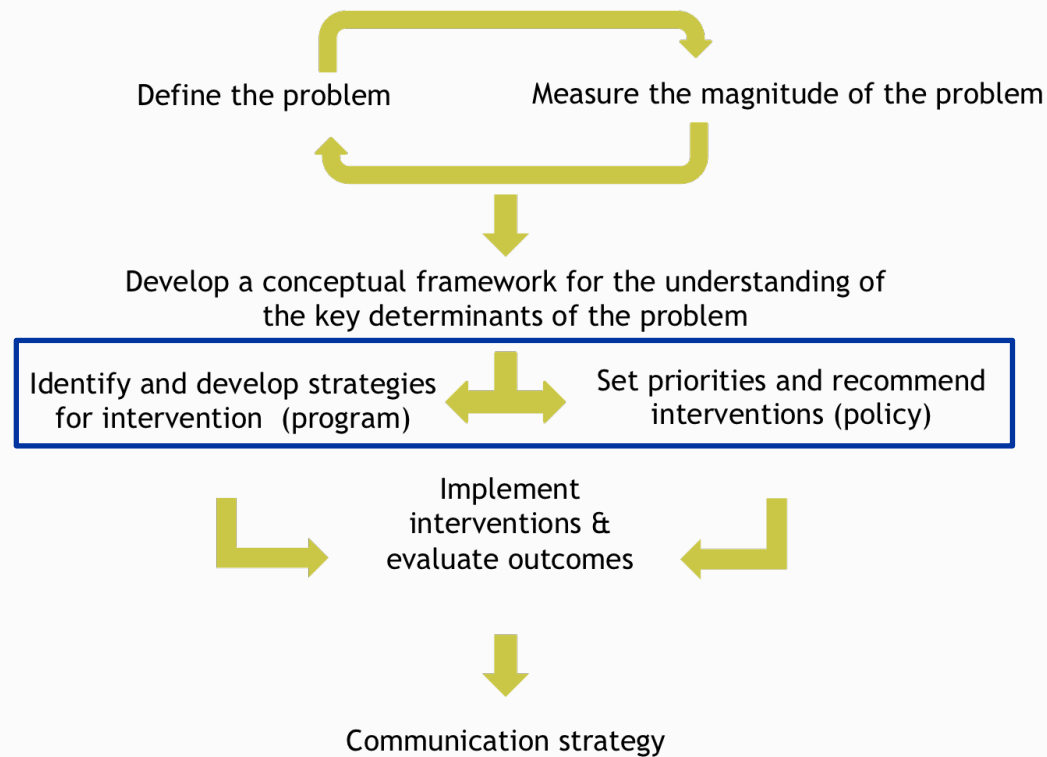
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“A goal without a plan is just a wish.”

*~Unknown*



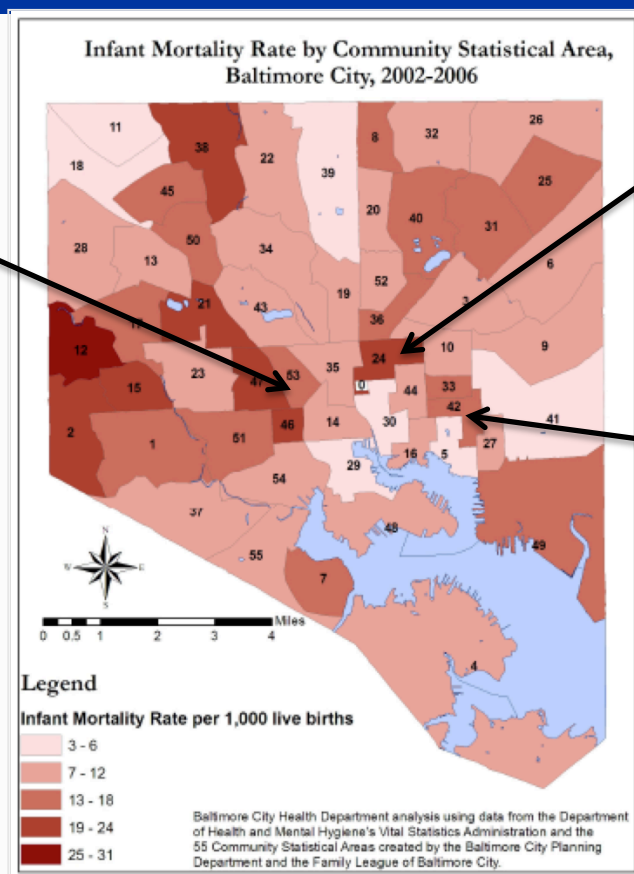
# Problem Solving Methodology



## Epidemiologists Inform Policy & Program

- How does one create or select a policy or program to solve the stated problem?
  - ▶ Analysis of person, place, time
    - Whom will you target for intervention?
  - ▶ Analysis of the determinants of the problem
    - How will you intervene?
- “Policy making is a complex political process that goes well beyond professional [public health] expertise alone.” ~Guyer

## Infant Mortality Rate in Baltimore City



Upton/Druid Heights  
14.3/1,000 live births

Greenmount East  
21.3/1,000 live births

Patterson Park North & East  
12.6/1,000 live births

Source: Baltimore City Dept of Health

## Epidemiologists Inform Policy & Program

*Table: Leading Cause of Infant Death in 2006, Baltimore City and Maryland*

| Cause                                                                      | Balt Rank | Balt Number | Balt Rate* | Balt % | MD Rank | MD Number | MD Rate* | MD % |
|----------------------------------------------------------------------------|-----------|-------------|------------|--------|---------|-----------|----------|------|
| Disorders relating to short gestation and unspecified low birth weight     | 1         | 31          | 317.7      | 26     | 1       | 123       | 158.9    | 20   |
| Sudden Infant Death Syndrome                                               | 2         | 20          | 205.0      | 17     | 3       | 67        | 86.5     | 11   |
| Congenital Abnormalities                                                   | 3         | 12          | 123.0      | 10     | 2       | 110       | 142.1    | 18   |
| Newborn affected by maternal complications of pregnancy                    | 4         | 10          | 102.5      | 8      | 4       | 46        | 59.4     | 8    |
| Bacterial sepsis of newborn                                                | 5         | 5           | 51.2       | 4      | 6       | 25        | 32.3     | 4    |
| Newborn affected by maternal complications of placenta, cord, and membrane | 6         | 4           | 41.0       | 3      | 5       | 34        | 43.9     | 6    |
| Newborn affected by other complications of labor and delivery              | 7         | 4           | 41.0       | 3      | -       | -         | -        | -    |
| Respiratory distress of newborn                                            | 8         | 4           | 41.0       | 3      | 7       | 23        | 29.7     | 4    |
| Neonatal hemorrhage                                                        | 9         | 3           | 30.7       | 3      | 8       | 15        | 19.4     | 2    |
| Pulmonary hemorrhage originating in the perinatal period                   | 10        | 3           | 30.7       | 3      | -       | -         | -        | -    |
| All causes (residual)                                                      |           | 25          | 256.2      | 21     | -       | -         | -        | -    |
| All Causes                                                                 |           | 121         | 1240.1     | 100    |         | 615       | 794.3    | 100  |

Source: DHMH Vital Statistics Administration. \*Rate per 100,000 live born infants.

## The interventions 1

- B'More for Healthy Babies
  - ▶ Main partners: Baltimore City Health Dept. and the Family League of Baltimore City
  - ▶ “Our hope is that all of Baltimore’s babies are born at a healthy weight, full term, and ready to thrive in healthy families.”

**WORK IT OUT >>**

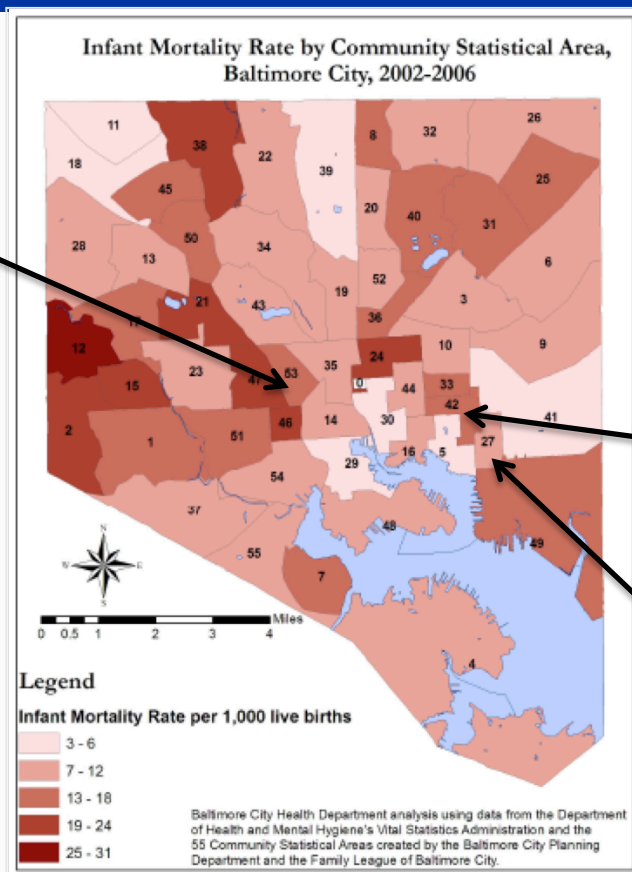
- [www.healthybabiesbaltimore.com](http://www.healthybabiesbaltimore.com)

## The interventions 2

- B'More for Healthy Babies
  - ▶ Accidental death and suffocation, SIDS
    - 2010 Safe Sleep intervention
    - 2011 Dad's Campaign (tobacco smoke)
  - ▶ Short gestation, low birth rate
    - 2012 B'More Fit for Healthy Babies
  - ▶ Multiple proximal causes
    - Baby Basics (ongoing)
    - Moms Clubs (ongoing)
    - Home visiting programs (ongoing)
    - Rattle & Role (annual)

## Infant Mortality Rate in Baltimore City

Upton/Druid Heights  
14.3/1,000 live births



Patterson Park North & East  
12.6/1,000 live births

Highlandtown  
8.1/1,000 live births

Source: Baltimore City Dept of Health

## Epidemiologists Inform Policy & Program

- How is one policy or program better than another?
- Strength of the evidence supporting/refuting each option
  - ▶ Ensure the policy/program has targeted the affected population (person, place)
  - ▶ Ensure the policy/program targets a modifiable determinant
  - ▶ Ensure the policy/program effect can be measured



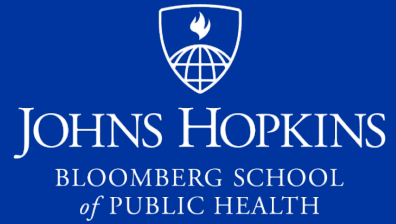
# Problem Solving Methodology

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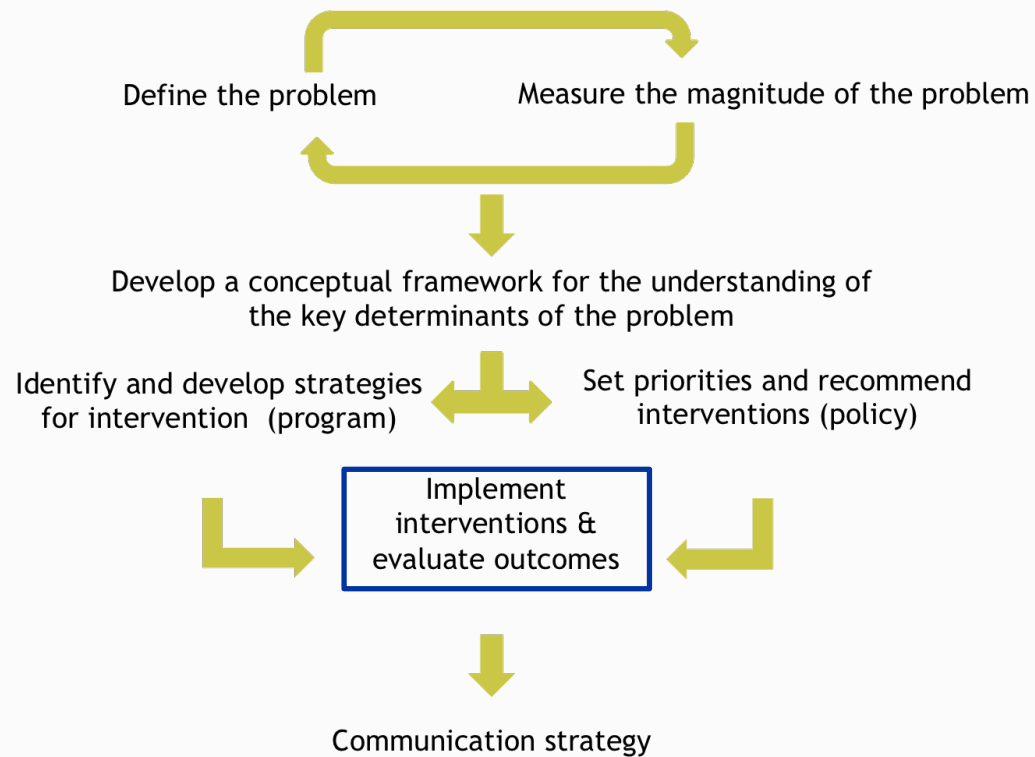
## Actions and Evaluations

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*"It always seems impossible until it's done."*

*~Nelson Mandela (1918-2013)*

# Problem Solving Methodology



## Epidemiologists monitor and evaluate 1

- The epidemiologist is heavily involved with monitoring and evaluation of the policy(ies)/program(s)
- What outcome will you be monitoring?
- What is the goal of the program?
  - ▶ Should be expressed in quantitative terms of the outcome being monitored for evaluation purposes
- How will we monitor the outcome?
  - ▶ A plan or a system to monitor the policy or program should be in place prior to implementation

## Epidemiologists monitor and evaluate 2

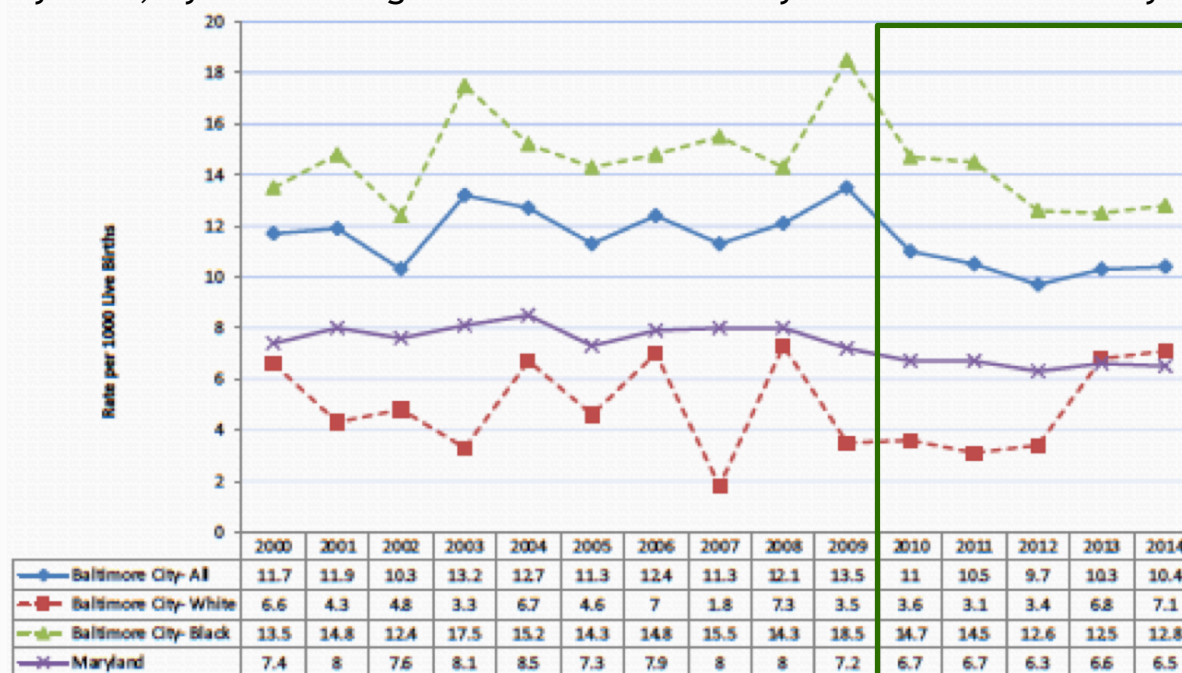
- The epidemiologist is most involved with monitoring and evaluation of the policy(ies)/program(s)
- How will we know if our policy/program was successful?
  - ▶ Comprehensive evaluation should occur after every policy or program

**WORK IT OUT >>**

- [www.healthybabiesbaltimore.com](http://www.healthybabiesbaltimore.com)

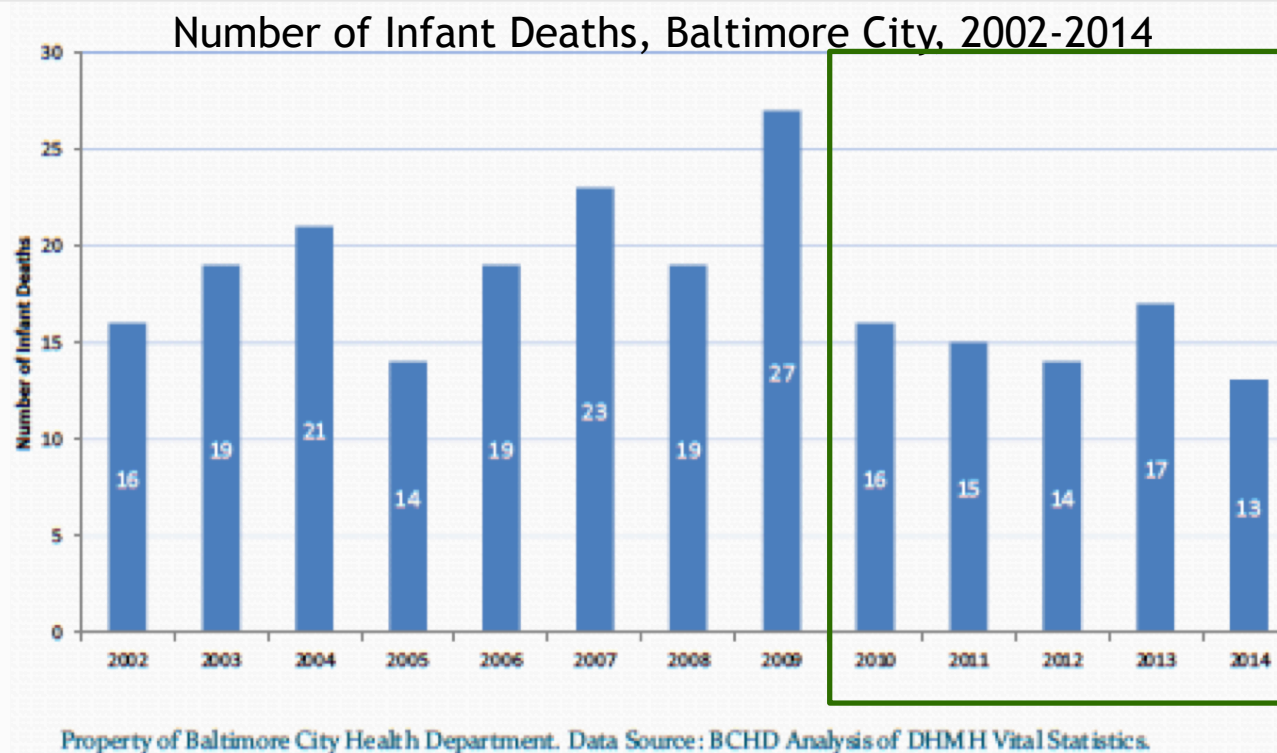
# The evaluation 1

Infant Mortality Rate, by race among those in Baltimore City and the state of Maryland, 2000-2014



Property of Baltimore City Health Department. Data Source: BCHD Analysis of DHMH Vital Statistics.

## The evaluation 2



Source: [http://www.healthybabiesbaltimore.com/uploads/files/Birth%20Profile%202010\\_14\\_general%20public%205\\_17\\_2016.pdf](http://www.healthybabiesbaltimore.com/uploads/files/Birth%20Profile%202010_14_general%20public%205_17_2016.pdf)

# Problem Solving Methodology

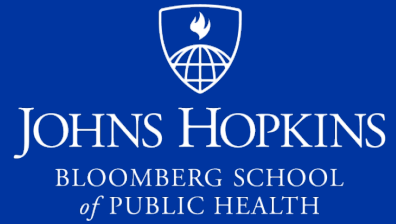
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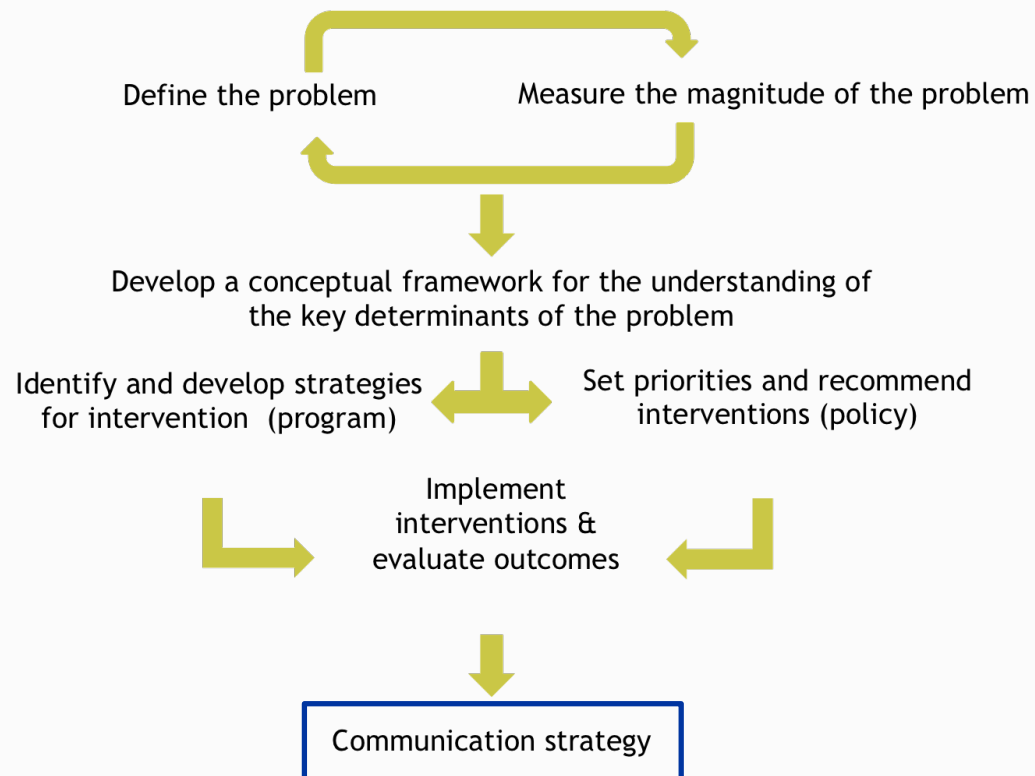
## Communication

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“The single biggest problem in communication is the illusion that it has taken place.”

*~George Bernard Shaw (1856-1950)*

# Problem Solving Methodology



## Epidemiologists inform communication

- Communication is necessary for disseminating results
  - ▶ Interventions that occur should be noted in future routine health assessments
- Communication is dependent upon
  - ▶ The message you are trying to send on behalf of the data
  - ▶ The audience
- Mechanisms of communication
  - ▶ Visual
  - ▶ Written
  - ▶ Oral

# The communication

The screenshot displays the website for B'more for Healthy Babies. The header includes the organization's logo, a navigation menu with links like 'About BHB', 'Our Initiatives', 'Moms Clubs', 'Home Visiting', 'Community Programs', 'News & Events', 'Parents & Caregivers', and 'Provider Toolkits'. A search bar is located in the top right corner. The main content area is titled 'Infant Mortality Statistics and Research' and features a sidebar with links to 'About BHB', 'Infant Mortality Statistics and Research', 'Vision & Strategy', 'Task Forces', 'Partners', 'BHB Leadership', 'BHB Team', 'Jobs & Internships', and 'Contact'. The main text section, titled '2014 INFANT MORTALITY DATA:', provides a detailed overview of the infant mortality rate in Baltimore City, comparing it to the United States and Maryland, and highlighting the impact of the B'more for Healthy Babies initiative. It also includes a list of 'Key Statistics' and a link to 'BCHD's Profile of Maternal and Infant Characteristics, Baltimore City, 2010-2014'. The footer contains a 'SIGN UP FOR UPDATES' button, a 'DONATE A CRIB' button, and a section for 'IMPORTANT LINKS' including a link to quit smoking.

Home :: About BHB :: Infant Mortality Statistics and Research

## Infant Mortality Statistics and Research

**2014 INFANT MORTALITY DATA:**

In 2009, the year B'more for Healthy Babies launched, Baltimore City had the fourth highest infant mortality rate (IMR) in the United States. The City's IMR has since fallen by 24%—from 13.5 deaths per 1,000 live births in 2009 to 10.4 in 2014, the latest year for which data is available. Although we cannot make a direct correlation between the initiation of B'more for Healthy Babies and the decrease in the IMR, the fact that we see this decrease is quite promising that the work is making a positive impact on maternal and infant health. However, Baltimore City's IMR continues to be higher than the Maryland IMR, and racial disparities, though shrinking, still exist.

**Key Statistics**

- In 2014, a total of 92 infants died, resulting in an infant mortality rate of 10.4 infant deaths per 1,000 live births
- Sleep-related infant deaths in Baltimore City have decreased by more than 20% since 2009 (from 27 deaths in 2009 to 13 deaths in 2014), which has greatly contributed to the decrease in the IMR
- In 2014, a black infant was about 2 times more likely to die than a white infant, resulting in 54 excess black infant deaths\*
- In 2014, Baltimore City's infant mortality rate was 1.6 times greater than Maryland's rate, resulting in 34 excess infant deaths\*

*\*Excess infant deaths are those that would have been prevented if the infant mortality rates were equal for both groups being compared*

BCHD's Profile of Maternal and Infant Characteristics, Baltimore City, 2010-2014

**Maryland Vital Statistics Infant Mortality Reports:**

- 2014 Maryland Infant Mortality Report
- 2013 Maryland Infant Mortality Report
- 2012 Maryland Infant Mortality Report

Source: <http://healthybabiesbaltimore.com/about-bhb/infant-mortality-statistics-and-research>